

Curriculum Vitae

Name: Maneesh Punetha

Date of Birth: September 01, 1989

Current Position: Assistant Professor,
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Research Gate: https://www.researchgate.net/profile/Maneesh_Punetha

Area of interest: Clean Energy, Nuclear Engineering, Nuclear Thermal Hydraulics, Hydrogen Storage and Safety, and Heat Transfer.



Academic Qualifications

Ph. D. – Mechanical Engineering – **Indian Institute of Technology Kanpur**, Uttar Pradesh, India 2014-2020 (**CGPA: 8.67** and **Outstanding Ph.D. Thesis Award in Mechanical Engineering - 2020, IIT Kanpur**)

Thesis Title: *Containment Thermal Hydraulic Studies towards Understanding Post-Severe Nuclear Accident Scenarios*

- **M. Tech.** – Thermal System Design – **Sardar Vallabhbhai National Institute of Technology, Surat**, India, 2012-2014 (**CGPA: 9.65**)

Thesis Title: *Analysis of Dispersion of Heated Effluent in Lake*

- **B. Tech.** – Mechanical Engineering – **Government College of Engineering Amravati**, Maharashtra, India, 2007-2011 (**CGPA: 8.45**)

Project Title: *Optimization of Number of Turns of Closed Loop Pulsating Heat Pipe (CLPHP).*

Post- Ph. D. Research Experience

- **December 2023 - Present** as Assistant Professor, School of Mechanical Sciences, IIT Bhubaneswar, Odisha, India
- **June 2021 - December 2023** as Postdoctoral Fellow, Division of Nuclear Power Safety, School of Engineering Sciences, KTH, Stockholm, Sweden (Supervisor: Prof. Sevostian Bechta)
- **August 2020 - June 2021** as Senior Research Fellow, Department of Mechanical Engineering, IIT Kanpur, Kanpur, India (Supervisor: Prof. Sameer Khandekar)

Industrial Experience

- **July 2011 - August 2012**, as Assistant Manager (Health, Safety and Environment) in **Essar Steel India Limited, Hazira**, Gujarat; attained Six Sigma Yellow Belt Certification.

Teaching Experience

- Teacher (Indian Institute of Technology Bhubaneswar, Odisha, India)
 - ME2L501 Elements of Mechanical Engineering
 - ME6L112 Two Phase Flow Heat Transfer
 - ME3L501 Applied Thermodynamics
 - ME6L162 Nuclear Power Generation and Safety
 - ME1P001 / CE1P001 Introduction to Manufacturing Processes (Laboratory)
- Teacher (KTH Royal Institute of Technology, Stockholm, Sweden)
 - FSH3001 HT21-1 Nuclear Power Safety Engineering Project
 - SH1015 HT21 Applied Modern Physics (Project)
- Teaching Assistant (Indian Institute of Technology Kanpur, Kanpur, UP, India)
 - Liquid-Vapour Phase-Change Phenomena
 - Refrigeration and Air Conditioning.

Sponsored Projects:

- Thermal Hydraulic Study for Passive Safety of Small Modular Reactor (SMR) Containment (PI: Maneesh Punetha), Amount- Rs. 2.394 Million INR, Sponsoring Agency- SRIC, IIT Bhubaneswar, Odisha, India, (Ongoing).

Academic Achievements/Fellowships

- Received “**Outstanding Ph. D. Thesis Award in Mechanical Engineering - 2020**” on 53rd Convocation of IIT Kanpur.
- “Student Best Poster Award” for 'Effect of surface inclination on film condensation heat transfer in the presence of non-condensable gases' at 27th International Conference on Nuclear Engineering (ICONE27), Ibaraki, Japan, May 19-24, 2019.
- International travel grant from IIT Kanpur for attending the 27th International Conference on Nuclear Engineering (ICONE27), May 19-24, 2019 at Tsukuba International Congress Center, Tsukuba, Ibaraki, Japan.
- Received higher education scholarship from MHRD, Government of India to pursue Ph.D. at Indian Institute of Technology Kanpur, Kanpur (U. P.), India (2014-2019) and M.Tech. at Sardar Vallabhbhai National Institute of Technology, Surat (Gujarat), India (2012-2014).
- Awarded "Times of India Merit Scholarship" in M. Tech for academic year of 2012-13.
- Awarded Merit scholarship by Army Welfare Education Society (AWES), New Delhi, India in all four years of B. Tech from 2007-2011.

List of Publications

1. Punetha M., Yadav M. K., Jain S., Khandekar S., and Sharma P. K., Thermal-Hydraulic Test Facility for Nuclear Reactor Containment: Engineering Design Methodology and Benchmarking, Progress in Nuclear Energy, Vol. 138, pp. 1-20, August 2021. DOI: [10.1016/j.pnucene.2021.103837](https://doi.org/10.1016/j.pnucene.2021.103837) **(Web of Science: Q1)**
2. Punetha M., Yadav M. K., Khandekar S., Sharma P. K., and Ganju S., Intrinsic Transport and Combustion Issues of Steam-Air-Hydrogen Mixtures in Nuclear Containments, International Journal of Hydrogen Energy, Vol. 45 (4), pp. 3340-3371, 2020. DOI: [10.1016/j.ijhydene.2019.11.179](https://doi.org/10.1016/j.ijhydene.2019.11.179) **(Web of Science: Q1)**
3. Punetha M., Choudhary A., and Khandekar S., Stratification and Mixing Dynamics of Helium in an Air-Filled Confined Enclosure, International Journal of Hydrogen Energy, Vol. 43 (42), pp. 19792-19809, 2018. DOI: [10.1016/j.ijhydene.2018.08.168](https://doi.org/10.1016/j.ijhydene.2018.08.168) **(Web of Science: Q1)**
4. Punetha M., and Khandekar S., A CFD based Modeling Approach for Predicting Steam Condensation in the Presence of Non-condensable Gases, Nuclear Engineering and Design, Vol. 324, pp. 280-296, 2017. DOI: [10.1016/j.nucengdes.2017.09.007](https://doi.org/10.1016/j.nucengdes.2017.09.007). **(Web of Science: Q1)**
5. Fang D., Deng Y., Xiang Y., Punetha M., Zhao L., and Ma W., Experimental and numerical studies on spray cooling of a downward-facing surface under partial coverage of multi-nozzle sprays, Nuclear Engineering and Design, Vol. 428, pp. 1-16, November 2024. DOI: [10.1016/j.nucengdes.2024.113525](https://doi.org/10.1016/j.nucengdes.2024.113525)
6. Zhao L., Punetha M., Ma W., Bechta S., Isaksson P., Lomperski S. W., Farmer M. T., and Licht J. R., Application of moving particle semi-implicit method on simulating melt spreading within OECD/ROSAU project, Nuclear Engineering and Design, Vol. 427, pp. 1-11, October 2024. DOI: [10.1016/j.nucengdes.2024.113447](https://doi.org/10.1016/j.nucengdes.2024.113447)
7. Zhao L., Punetha M., Ma W., Konovalenko A., and Bechta S., Simulation of Melt Spreading over Dry Substrates with the Moving Particle Semi-implicit Method, Nuclear Engineering and Design, Vol. 405, pp. 1-12, April 2023. **(Web of Science: Q1)**
8. Yadav M. K., Punetha M., Bhanawat A., Khandekar S., and Sharma P. K., Steam Condensation Heat Transfer during Initial Blow-down Period of a Severe Nuclear Accident, ASME Journal of Nuclear Engineering and Radiation Science, Vol. 6 (4), pp. 1-9, October 2020. DOI: [10.1115/1.4046910](https://doi.org/10.1115/1.4046910) **(Web of Science: Q3)**
9. Bhanawat A., Yadav M. K., Punetha M., Khandekar S., and Sharma P. K., Effect of Surface Inclination on Filmwise Condensation Heat Transfer During Flow of Steam–Air Mixtures, ASME Journal of Thermal Science and Engineering Applications, Vol. 12 (4), pp. 1-12, August 2020. DOI: [10.1115/1.4046867](https://doi.org/10.1115/1.4046867) **(Web of Science: Q1)**

Book Chapters

1. Yadav M.K., Punetha M., Bhanawat A., Khandekar S., and Muralidhar K., Measurement of Condensation Heat Transfer, in 'Drop Dynamics and Dropwise Condensation on Textured Surfaces', Mechanical Engineering Series, Springer, 2020. DOI: [2019-20 10.1007/978-3-030-48461-3_13](https://doi.org/10.1007/978-3-030-48461-3_13)
2. Punetha M., Thermal Pollution: Mathematical Modelling and Analysis, in 'Environmental Contaminants', Energy, Environment and Sustainability, Springer, Singapore, 2018. DOI: [10.1007/978-981-10-7332-8_18](https://doi.org/10.1007/978-981-10-7332-8_18)

Peer Reviewed Conference Proceedings (published/presented)

1. Pathy P. P., and Punetha M., Thermal Hydraulic Study for Passive Safety of Small Modular Reactor (SMR) Containment. 11th International and 51st National Conference on Fluid Mechanics and Fluid Power (FMFP - 2024), AMU Aligarh, India December 21-23, 2024.
2. Punetha, M., and Khandekar S., (Invited) Containment Thermal Hydraulics Study for a Nuclear Power Plant Disasters. 4th Biennial International Conference on Future Learning Aspects of Mechanical Engineering (FLAME - 2024), Amity University, Noida, Uttar Pradesh, 31st July to 2nd Aug 2024
3. Punetha, M., Komlev, A., Konlvalenko, A., and Bechta, S., Thermo-hydrodynamic of Corium Simulant Melt Spreading: A small scale model study. '11th International Conference on Multiphase Flow (ICMF-2023)', Kobe, Japan. 2-7, April 2023
4. Punetha M., Kulkarni S., Yadav M.K., and Khandekar S., A CFD Study on Coupled Issues of Hydrogen Distribution and Steam Condensation Inside Thermal Hydraulic Test facility for Containment (THYCON), 25th National and 3rd International ISHMT-ASTFE Heat and Mass Transfer Conference, IIT Roorkee, Uttarakhand, India, December 28-31st, 2019
5. Punetha M., Yadav M.K., Bhanawat A., and Khandekar, S., Steam Condensation Heat Transfer inside Reactor Containment during the Initial Transient of a Severe Accident, Proceedings of 27th International Conference on Nuclear Engineering (ICONE27), Tsukuba, Ibaraki, Japan, May 18-24, 2019. DOI: [10.1299/jsmeicone.2019.27.2166](https://doi.org/10.1299/jsmeicone.2019.27.2166)
6. Bhanawat A., Punetha M., Yadav M.K., and Khandekar, S., Effect of Surface Inclination on Film Condensation Heat Transfer in the Presence of Air, Proceedings of 27th International Conference on Nuclear Engineering, (ICONE27), Tsukuba, Ibaraki, Japan, May 18-24, 2019. DOI: [10.1299/jsmeicone.2019.27.2133](https://doi.org/10.1299/jsmeicone.2019.27.2133)
7. Kulkarni S., Punetha M., Choudhary A., and Khandekar S., Effect of Stratification and Natural Circulation on Steam Condensation in Presence of Non-Condensable Gases, Proceedings of 5th International Conference on Computational Methods for Thermal Problems (ThermaComp - 2018), IISc Bangalore, Karnataka, India, pp. 480-483, July 9-11, 2018. ISSN: [23055995](https://doi.org/10.1299/jsmeicone.2019.27.2133)
8. Punetha M., Choudhary A., Khandekar S. and Sharma P., Helium Stratification and Mixing Studies in a Fully Enclosed Chamber, 24th National Heat and Mass Transfer Conference and 2nd International ISHMT-ASTFE Heat and Mass Transfer Conference, BITS Hyderabad, Telangana, India, December 27-30, 2017.
9. Punetha M., and Khandekar S., Study of Film-wise Condensation inside Closed Containment using Wall Condensation Model (WCM), 6th International and 43th National Conference on Fluid Mechanics and Fluid Power (FMFP2016), Motilal Nehru National Institute of Technology, Allahabad, Uttar Pradesh, India, December 15-17, 2016.
10. Punetha M., Thaker, J. P., and Banerjee J., Experimental and Numerical Analysis of Dispersion of Heated Effluent from Power Plants, 5th International and 41th National Conference on Fluid Mechanics and Fluid Power (FMFP2014), Indian Institute of Technology Kanpur, Uttar Pradesh, India, December 12-17, 2014.
11. Punetha M., Roopchandani C., and Banerjee J., Analysis for dispersion of thermal effluent from Thermonuclear Power Plant, 40th National Conference on Fluid Mechanics and Fluid Power (FMFP2013), National Institute of Technology Hamirpur, Himachal Pradesh, India, December 12-14, 2013.

Detailed Research Work

Postdoctoral research

- **Project 1:** *Reduction of Severe Accident Uncertainties (ROSAU) for Reactor Core-Debris Cooling in Shallow Containment Cavity* (Sponsor: OECD Nuclear Energy Agency)
 - The objective of the work is to study the hydrodynamic and thermal interaction of corium, a highly radioactive material, during its spreading over containment cavity floor in order to understand the phenomenology and develop numerical codes. The cavity has a shallow water pool, which simulates a Nordic PWR.
 - A new facility known as Co-rium Simulant Melt Underwater Spreading (CoSMUS) is developed with advanced measurement capabilities. Experiments were conducted based on project goals.
- **Project 2:** *In-Vessel Melt Retention (IVMR) for Reactor Core-Melt Cooling in Reactor vessel lower head* (Sponsor: Swedish Radiation Safety Authority).
 - The research is being conducted in two parts (a) Radiation Modelling of In-vessel Melt Coolability, (b) Spray cooling for In-Vessel Melt Retention (IVMR).
 - The goal of this research is to increase confidence in the IVR strategy and guide modification in existing reactors and new designs.

Ph.D. Research

- **Project 1:** *Studies on heat transfer during condensation of steam-hydrogen mixtures inside closed containment*, funded by Bhabha Atomic Research Center (BARC), Mumbai, India. Duration: October 2015 - March 2020 (Also the main work of PhD thesis).
 - Designed, fabricated, and commissioned a single-compartment large scale Thermal-HYdraulic test facility for CONtainment (THYCON) to simulate Fukushima Daichi like of accident in Indian BWR. Fixed instrumentation with advanced measurement capability.
 - Performed calibration and benchmarking of a complex online mass-spectrometry system (Hiden Analytical® make with 20 sampling ports) to estimate spatial composition of mixture gasses (steam, air and hydrogen), which have largely different properties.
 - Implemented an inverse technique based high heat flux measurement system for adverse situations.
 - Prepared project progress report/final report and mentoring undergraduate/ graduate students.
 - Published three articles (in section "List of Publications" #1, #2 and #3).
- **Project 2:** *Local heat transfer coefficient during film condensation of steam hydrogen mixtures*, funded by Board of Research in Nuclear Sciences (BRNS), Mumbai, India. Duration: April 2015 - March 2018. The work carried out during this project includes:
 - Assistance in design, fabrication and instrumentations of the experimental set-up (CONFLO facility) for generic condensation studies in presence of non-condensable gasses.
 - Experiments conducted to study steam condensation heat transfer in the presence of only air and air-helium gases, at all stages of a severe nuclear accident progression.
 - Published two articles (in section "List of Publications" #4 and #5).

PhD Thesis Supervision: 1 (completed) + 2 (ongoing)

- Ms. Lu Zhao. Title of the Thesis: Simulation of Melt Infiltration and Spreading using Moving Particle Semi-implicit Method, [Supervising jointly with Prof. Sevostian Bechta, Prof. Weimin Ma, Dr. Alexander Konovalenko. **Status:** Completed].
- Mr. Prabhu Prasad Pathy. Title of the Thesis: Thermal hydraulics study of Small Modular Reactor (SMR) containment in the event of severe accident, [Sole Supervisor, **Status:** Ongoing]
- Mr. Mohammad Monzur Hossain Khan*. Title of the Thesis: Radiation Modelling of Reactor Pressure Vessel Lower Head, [Supervising jointly with Prof. Sevostian Bechta, Dr. Andrei Komlev. **Status:** Ongoing].

M.Tech. Thesis Supervision: 0 (completed) + 3 (ongoing)

- Mr. Latesh Govindrao Munde (Roll No. 23TSO6012), Title of the Thesis: Thermo-hydrodynamic study of corium in reactor cavity, [Sole supervisor, Status: Ongoing].
- Mr. Theeda Ravindra (Roll No. 23TSO6005), Title of the Thesis: Effect of radiation on reactor pressure vessel's molten pool, [Sole Supervisor, Status: Ongoing].
- Mr. Kanad Mitra (Roll No. 23TSO6016), Title of the Thesis: Minimization of boiloff losses for liquid hydrogen storage, [Sole Supervisor, Status: Ongoing].

Invited Speaker:

- Technical Session 'In the loving memory of Prof. Sameer Khandekar', FMFP-2024, 21st December 2024, Z.H. College of Engineering & Technology, Aligarh Muslim University, Aligarh (U.P), India.
- Topic: Rector Core Melt Relocation in a Severe Nuclear Accident Amity School of Engineering Technology, 20th December 2024, Amity University, Noida (U.P.), India.
- Session on 'Technology in Nuclear and Space' on National Science Day, 28th February 2024, Manas College of Science Technology and Management, Pithoragarh (Uttarakhand), India.
- 'Underwater Simulant-Corium Melt Spreading' at International Workshop on Fuel-Coolant Interaction and Steam explosion, IRSN Workshop on Fuel Coolant Interaction, Fontenay-aux-Roses (France).

Reviewer of Journals:

Nuclear Engineering and Design (Elsevier), International Journal of Hydrogen Energy (Elsevier), International Communications in Heat and Mass Transfer (Elsevier), International Journal of Heat and Mass Transfer (Elsevier), ASME Journal of Thermal Science and Engineering Application (ASME), ASME Journal of Heat Transfer (ASME), Water Quality Research Journal (IWA Publishing).

Other Experience

- Faculty advisor of B.Tech. - 24 Mechanical Engineering Batch of IIT Bhubaneswar, India from July 2024 - Ongoing
- Co-PIC: Heat Transfer Laboratory, and Thermofluid-II laboratory at SMS, IIT Bhubaneswar, India (March, 2024 - continuing)

- Institute Representative (IR) at JEE (Advanced) 2024 (25-26 May 2024)
- Maintained website of Mechanical Engineering Department, IIT Kanpur from July 2015-December 2019.

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